

Mitigating Adversarial Vulnerabilities in Brain Tumor Imaging: Threat Analysis and Defense Strategies

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Abstract—Medical image analysis using deep learning has made tremendous progress, but the robustness of such systems in realclinical settings is still hampered by model instability and adversarial sensitivity. This paper presents an efficient ensemble-motivated framework for MRI-based brain tumor prediction, incorporating three state-of-the-art networks (including InceptionV3, MobileNetV2 and ResNet50) to enhance diagnosis correctness and prognosis stability. In order to compensate the lack of dataset, we used heavy augmentations such as adding Gaussian noise and rotation to between fivefold of the amount in training data and obtained the improved generalization. Every model was fine-tuned on augmented dataset, and their probabilistic predictions were aggregated via soft-voting ensemble learning to achieve more consistent and accurate results than any single model. Apart from general classification, we focus on adversarial threats to medical imaging systems in this work. We used four popular adversarial attack algorithms—FGSM, BIM, PGD and CW to analyze how easily our model could be fooled by the imperceptible perturbations that are able to corrupt clinical judgment. To defend against these threats, adversarial training and ensemble architectural diversity were employed as defenses. Experimental results show adding the proposed technique increases this accuracy to 92% and our particularly higher recall in tumours lead to a lower risk of false-negative diagnoses. In addition, the ensemble is more robust under adversarial perturbations than each individual model. In general, this work emphasizes the importance of combining high-performance classification and adversarial robustness for safe, reliable, as well as trustworthy AI-based brain tumor detection in clinical care.

Index Terms—Deep Learning, Brain Tumor, Health care, Image processing, Classification

I. INTRODUCTION

Recent advances in machine learning (ML) and artificial intelligence (AI) applied to image processing have resulted in significant strides across several dimensions including facial recognition, autonomous vehicles, medical diagnostics. Deep learning is a branch of AI which models the way human brain processes information by extracting features and making decisions automatically with multiple layers of structures in

neural network [22]. There are many powerful techniques available to deep learning models in the case of complex image classification task, because they can learn hierarchical features representation [15], [24].

Regardless of being benign or malignant, brain tumors are a significant threat to human health because of their unpredictable growth patterns and extremely diversified appearance. Size, shape and location of tumor vary greatly which make the early detection and accurate diagnosis difficult [14]. Magnetic Resonance Imaging (MRI) is known to be gold standard for non-invasive brain examination due to capabilities in superior visualization of soft tissue structures as compared to CT [20]. Deep learning-based methods can learn to automatically capture the discriminative characteristics of MRI scans that are helpful for clinicians when diagnosing and classifying brain tumors [6].

Although deep learning has been successful for medical image analysis, it is inherently susceptible to adversarial attacks — small, carefully constructed perturbations that are imperceptible to humans but can fool a neural network into producing an incorrect prediction [11]. These attacks raise critical patient safety concerns in medical domains, where misclassification could result in delayed or erroneous treatment. Therefore, it is important to have a clear understanding of adversarial threats and design strong defense mechanisms for securing AI-based medical systems.

With these difficulties in mind, the present paper explores a deep learning and adversarial robustness-oriented methodology for brain tumor detection via MRI images. We integrate three advanced CNN models (InceptionV3, MobileNetV2 and ResNet50) to exploit the complementary power for improved diagnostic decisions. Our key contributions consist in the following:

- **Improved Dataset Preparation:** We curated MRI im-

ages of healthy and tumor-affected brain according to publicly available datasets, and increased its dimensionality by five-fold, using rotation-based augmentation and gaussian noise. This enhanced diversity benefit the model in generalize to unknown data and is less prone to overfitting.

- **Reliability-enhanced Ensemble Learning:** Each Google Cloud API model (InceptionV3, MobileNetV2, ResNet50) was fine-tuned individually on the augmented dataset. Then they were aggregated in an ensemble manner, and the predictions are much more stable and accurate than any single model.
- **Optimized Deep Learning Architectures:** We utilized transfer learning as all three models' pretrained base layers are frozen, allowing efficient emergency training while maintaining meaningful feature representation.
- **Adversarial Robustness Analysis:** Due to increasingly threat of adversarial attacks in medical AI, we justified model robustness and examined defense strategies to kept reliability under potential disturbances.

Overall, this study endeavors to boost both diagnostics accuracy and safety for AI-based brain tumor detection systems by adopting state-of-the-art deep learning techniques, augmented data manipulation, and adversarial robustness considerations.

II. RELATED WORKS

Tumor detection and segmentation automation has witnessed extensive research since artificial intelligence (AI) has been proved to play a potential role in improving diagnostic accuracy and clinical decision making. Early solutions predominantly leveraged classical machine learning based on feature extraction, and handcrafted descriptors. Oliveira et al. offered an extensive review of feature extraction, selection and classification in pigmented lesion analysis with a perspective on how computer-aided diagnosis may best benefit from these methods [19]. Similarly, Abbas et al. used 350 MRI images with a multiclass SVM classifier to obtain 93% accuracy showing that traditional ML based methods are still useful when proper feature engineering is being performed [1]. We introduced structured feature transformation in our previous work methods using feature-grouped decision trees, showing how skillfully learned representations could improve the stability and accuracy of classification models [7]. Janney (2020 classification) employed adaptive k-means clustering for melanoma segmentation, then they classified lesions by KNN, SVM, MLP and Decision Tree and Random Forest Classifiers; this demonstrating the importance of classical algorithms on medical image analysis.

The transition from traditional machine learning to deep learning has greatly improved tumor classification and segmentation. Esteva et al. Dermatologist-level performance was previously reported by Esteva et al. 2017 dermatologist using a deep convolutional neural network (CNN) trained on more than 129,000 images. This pioneering work further promoted the application of CNN-based methods in more general medical

imaging setting. GANs have emerged recently also in MRI-based brain imaging. Recent survey showcased GANs for image to image translation, reconstruction, data-augmentation and segmentation [17]. GAN-based synthetic data may increase model robustness, although issues about image realism and patient safety persist.

Additionally, transformer family architectures have brought another wave to brain tumor segmentation. Wang et al. proposed TransBTS, which is a hybrid model of 3D CNNs and transformer modules designed to learn local and global spatial relationships, and demonstrated state-of-the-art performance on BraTS 2019 and 2020 datasets [27]. Hatamizadeh et al. introduced UNETR which fuses pure transformer encoder with U-Net style decoder achieving higher quality of volumetric MRI segmentation [9]. Building upon the transformer-based models, Cao et al. developed Swin-UNet that makes use of hierarchical shifted windows for the multi-scale features and superior to conventional CNN architectures [4]. All of these works together demonstrate the fast transition from CNN to transformer based models in medical image segmentation.

A number of researchers have also worked on establishing strong, generalisable baselines. Isensee et al. proposed nnU-Net that is a self-configuring pipeline for medical image segmentation which automatically adjusts the preprocessing steps, allows different training schedules and modifications to the network architecture based on dataset-specific parameters, leading to state-of-the-art performance in biomedical image segmentation [12]. Moreover, Ma et al. developed MedSAM, an ultralarge foundation model fine-tuned with 1.57 million medical images and masks, achieving anti-scrub segmentation transfer across multiple categorically different imaging modalities [3]. These foundational models are pulling medical AI in the direction of something more all-purpose and flexible.

Related work: There has been parallel study on federated learning (FL) to combat data sharing limitations from medical environments. Sheller et al. performed a comprehensive benchmark using the FeTS 2022 dataset and finds that FL can achieve close-to-centralized performance for preserving patient privacy across institutions [23]. Hybrid deep learning architectures are also explored—Albahri et al. used a Vision Transformer (ViT) in conjunction with the GRU for tumor classification using MRI data, and achieved competitive results over tumor subtypes [2].

Huge challenges such as BraTS are still driving innovation. The 2024 challenge further added post-treatment gliomas segmentation tasks, such as resection cavity delineation; moving participant towards real clinical scenario [3]. These tasks are dominated by not just the segmentation accuracy but also robustness in challenging imaging conditions with treatment-induced artifacts.

Generally speaking, the reviewed studies showed noticeable improvements in terms of both conventional and deep learning-based methods for BT and BS. But it has been less successful when it comes to adversarial attacks, for example. Further, guaranteeing the robustness of state-of-the-art architectures to adversarial perturbations becomes essential when aiming

forsafe clinical deployment. This gap informs the current study to leverage ensemble deep learning and adversarial robustness into an impactful reliable MRI-based brain tumordetection model.

III. METHODOLOGY

The proposed system integrates modern deep learning architectures with ensemble learning and adversarial robustness techniques to ensure accurate and secure MRI-based brain tumor detection. The complete workflow consists of the following major stages: (1) data acquisition and preprocessing, (2) dataset augmentation, (3) training of three pretrained CNN models, (4) ensemble fusion for improved prediction stability, (5) generation of adversarial examples to evaluate model vulnerability, and (6) application of defense strategies. This section provides a comprehensive and structured overview of each component. This emphasis on improving representation learning aligns with our earlier work on correlation-based feature extraction for enhancing tabular classification performance [13]. Although the present study focuses on deep visual representations, both approaches highlight the importance of feature quality in improving model generalization.

A. Data Acquisition and Preprocessing

The MRI dataset used in this study was collected from publicly available online repositories and contains both normal brain MRI scans and images showing tumor regions. Since the dataset sources vary, differences such as resolution, brightness, and orientation were harmonized through a preprocessing pipeline.

All images were resized based on the required input of the deep learning architectures: 224×224 for MobileNetV2 and ResNet50, and 299×299 for InceptionV3. Pixel intensities were normalized to the range $[0, 1]$ to stabilize gradient flow and reduce the influence of imaging variations such as brightness or scanner noise.

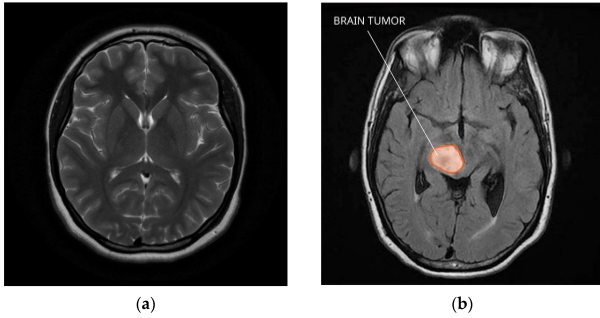


Fig. 1: Normal brain and tumor brain MRI dataset. (a) Normal MRI; (b) Tumor MRI.

Additional preprocessing steps included orientation correction and removal of unnecessary border regions to ensure that the model learned only meaningful tumor-related features.

B. Augmentation and Dataset Expansion

Medical datasets are generally limited in size due to patient privacy and data acquisition challenges. To address this

limitation, augmentation techniques were applied to increase dataset diversity by approximately fivefold. Two primary augmentation methods were used:

- **Rotation:** Images were rotated at fixed angles (e.g., 15° , 30° , 45°) to enhance the model's ability to generalize across different patient head orientations.
- **Gaussian Noise Injection:** Controlled Gaussian noise was added to simulate realistic MRI scanner variability, improving robustness to imaging artifacts.

These augmentations reduce overfitting and help the model learn invariant tumor features.

C. Training of Individual Deep Learning Models

Three pretrained convolutional neural networks—InceptionV3, MobileNetV2, and ResNet50—were fine-tuned for binary classification (tumor vs. non-tumor). Each architecture contributes uniquely to the final ensemble:

- **InceptionV3** [25]: Uses parallel convolutional filters of varying sizes to extract multi-scale information.
- **MobileNetV2** [21]: Lightweight architecture optimized through inverted residual blocks and depthwise separable convolutions.
- **ResNet50** [10]: Employs residual skip connections to avoid vanishing gradients in deeper layers.

The pretrained layers were frozen to retain previously learned general visual representations, while the top layers were retrained using our MRI dataset. The training used the Adam optimizer with a learning rate of 1×10^{-4} and binary cross-entropy loss.

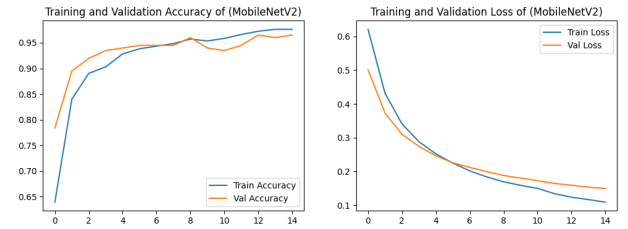


Fig. 2: MobileNetV2 performance: (a) Training/Validation accuracy; (b) Training/Validation loss.

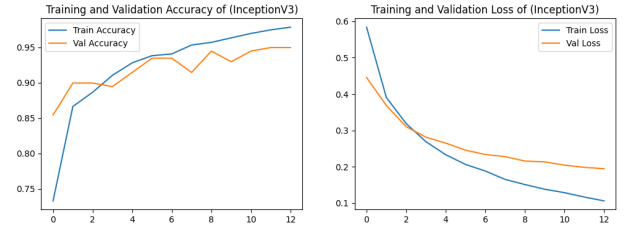


Fig. 3: InceptionV3 performance: (a) Training/Validation accuracy; (b) Training/Validation loss.

The complementary nature of these models motivated the ensemble design.

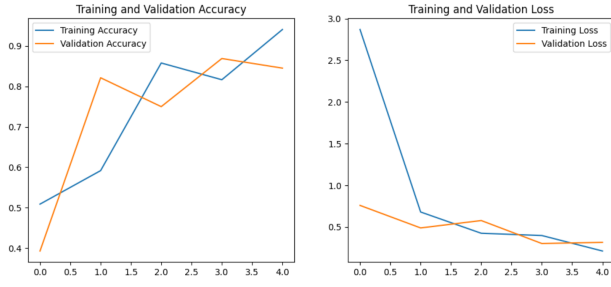


Fig. 4: ResNet50 performance: (a) Training/Validation accuracy; (b) Training/Validation loss.

D. Ensemble Learning Framework

Since each model learns different representations, combining their predictions enhances overall stability and accuracy. A soft voting (probability averaging) approach was used:

$$P_{\text{ensemble}} = \frac{P_{\text{IncV3}} + P_{\text{Mobile}} + P_{\text{ResNet}}}{3}$$

This approach improves:

- **Stability:** Reduces sensitivity to incorrect predictions from individual models.
- **Accuracy:** Aggregates diverse feature representations.
- **Adversarial Resistance:** Attacks targeting one model may not transfer effectively to the others.

E. Adversarial Attack Models

Deep learning classifiers are highly vulnerable to adversarial examples—images modified by carefully crafted, imperceptible perturbations.

1) *Fast Gradient Sign Method (FGSM)*: FGSM [8] is a single-step attack that perturbs the input image in the direction of the gradient of the loss function:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

FGSM is widely used due to its simplicity and efficiency.

2) *Basic Iterative Method (BIM)*: BIM [16] extends FGSM by applying it iteratively with small step size α :

$$x^{t+1} = \text{Clip}_{x, \epsilon}(x^t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x^t, y)))$$

This produces stronger perturbations while ensuring the final adversarial example remains within an ϵ -radius ball.

3) *Projected Gradient Descent (PGD)*: PGD [18] is considered one of the most powerful first-order adversarial attacks. It performs iterative gradient updates while projecting the result back into a valid perturbation space:

$$x^{t+1} = \Pi_{B_\epsilon(x)}(x^t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x^t, y)))$$

PGD is often treated as the “universal adversary” baseline for robustness evaluation.

4) *Carlini–Wagner (CW) Attack*: The CW attack [5] uses an optimization-based objective to find a minimal perturbation δ :

$$\min_{\delta} \|\delta\|_2 + c \cdot f(x + \delta)$$

CW is designed to bypass many defenses and is one of the most difficult attacks to mitigate.

F. Defense Strategies

Two defense mechanisms were integrated into the system:

- **Adversarial Training:** Clean images were mixed with FGSM and PGD adversarial examples during training. This exposes the model to adversarial patterns and strengthens robustness.
- **Ensemble Defense:** Since adversarial examples do not always transfer across architectures, the ensemble benefits from inherent defense through model diversity.

These defenses significantly reduce the effectiveness of both single-step and iterative attacks.

G. Attack–Defense Evaluation Pipeline

The complete evaluation process is summarized below:

- 1) Baseline classification performance is measured on clean MRI images.
- 2) FGSM, BIM, PGD, and CW adversarial examples are generated.
- 3) The ensemble is evaluated on adversarial images to quantify vulnerability.
- 4) Adversarial training is performed to enhance robustness.
- 5) Post-defense evaluation is conducted to measure improvement on both clean and adversarial samples.

This pipeline ensures the system is thoroughly assessed under realistic diagnostic and adversarial conditions.

IV. RESULT ANALYSIS

The performance of the proposed ensemble-based deep learning framework was evaluated using several key metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of how effectively the system distinguishes between tumor and non-tumor MRI images. In addition to clean-data evaluation, adversarial robustness was also analyzed to determine the stability of the model under malicious perturbations.

A. Overall Classification Performance

The ensemble model achieved an overall accuracy of 92%, demonstrating strong capability in differentiating tumor and non-tumor brain MRI scans. This accuracy reflects the benefit of combining InceptionV3, MobileNetV2, and ResNet50, where the ensemble averages out the weaknesses of individual models and enhances both stability and generalization.

The confusion matrix and classification metrics provide deeper insight into class-wise performance. For the non-tumor class (Class 0), the system achieved a precision of 93% and a recall of 91%. The high precision indicates that the model

makes few false positive predictions, whereas the strong recall suggests that the majority of non-tumor images were identified correctly.

For the tumor class (Class 1), the model achieved a precision of 91% and a recall of 94%. The high recall is especially important in medical diagnosis because it indicates the system’s ability to identify tumor cases with minimal false negatives. Missing a tumor can lead to delayed treatment and severe consequences; thus, the system’s focus on minimizing false negatives is clinically meaningful.

The F1-scores for both classes were balanced at 92%, demonstrating consistency across classes. The macro-average and weighted-average values both stood at 92%, indicating that the model maintained stable performance even if class distributions were slightly unbalanced.

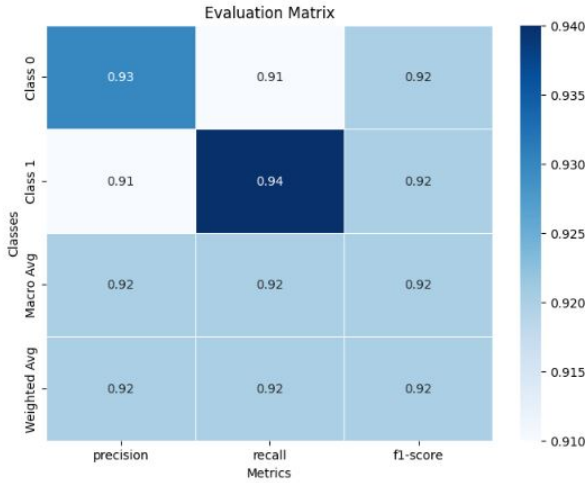


Fig. 5: Evaluation metrics for MobileNetV2: precision, recall, and F1-score for Class 0 and Class 1, along with macro and weighted averages.

B. Performance of Individual Models

Before combining the models into an ensemble, each network was evaluated independently. MobileNetV2, InceptionV3, and ResNet50 exhibited strong yet distinct performance characteristics:

- **MobileNetV2:** Lightweight and fast, showing stable convergence. Its accuracy stabilized at around 85–88% during validation.
- **InceptionV3:** Capable of multi-scale feature extraction, achieving up to 90.56% accuracy with moderate loss values.
- **ResNet50:** Produced the highest standalone accuracy (94.08%), effectively capturing deep hierarchical features.

Although ResNet50 individually achieved the highest accuracy, the ensemble outperformed all three models in terms of robustness and stability. The ensemble prediction smooths out fluctuations from individual models and improves generalization, especially in edge-case images.

C. Impact of the Ensemble Strategy

The ensemble classifier demonstrated improved consistency and reduced prediction variance. By averaging the softmax probabilities of all three models, the final predictions were less sensitive to misclassifications made by any individual architecture.

This improvement was particularly noticeable in MRI scans containing subtle or low-contrast tumor regions, where a single model might misinterpret patterns. The ensemble approach provided a balanced view by combining multi-scale, lightweight, and deep residual feature extraction capabilities.

D. Adversarial Robustness Evaluation

To examine the security and reliability of the system under adversarial manipulation, FGSM, BIM, PGD, and CW attacks were generated. These adversarial examples were crafted to mislead the model while keeping the perturbations visually imperceptible.

When the individual models were tested directly against adversarial samples, a significant drop in accuracy was observed, confirming the vulnerability of standard CNNs. However, the ensemble classifier demonstrated improved resistance due to reduced transferability of adversarial perturbations across different architectures.

Further robustness was achieved through adversarial training, during which the model was exposed to FGSM and PGD adversarial examples. After this training phase, the ensemble exhibited improved performance against both single-step and iterative attacks. The adversarially trained ensemble maintained significantly higher accuracy than any single model, reinforcing its suitability for real-world medical applications where adversarial threats may compromise patient safety.

E. Clinical Significance of Performance

The results highlight the system’s practical value in clinical settings:

- The high recall for the tumor class reduces the risk of missed diagnoses.
- Balanced performance across metrics ensures consistent results across diverse MRI samples.
- Improved robustness against adversarial attacks enhances security for AI-assisted diagnosis.

Overall, the combination of high accuracy, strong tumor-detection recall, and adversarial robustness indicates that the proposed system can provide reliable support in medical imaging workflows.

V. CONCLUSION

This study presented a comprehensive and secure deep learning framework for MRI-based brain tumor detection by integrating three complementary pretrained architectures—InceptionV3, MobileNetV2, and ResNet50—into an ensemble classifier. The ensemble approach effectively combined multi-scale feature extraction, computational efficiency, and deep residual learning, resulting in improved accuracy, stability, and generalization compared to individual models.

The system demonstrated strong performance on clean MRI scans, delivering high precision, recall, and overall diagnostic reliability, which highlights its potential as an assistive tool for radiologists in early tumor detection.

A key contribution of this work lies in its focus on adversarial robustness, an increasingly critical challenge in medical AI. Modern deep learning models are vulnerable to adversarial attacks, where carefully crafted perturbations can cause incorrect predictions while remaining visually imperceptible. To address this, four widely studied attack algorithms—FGSM, BIM, PGD, and CW—were implemented to evaluate the model’s susceptibility to malicious manipulation [5], [8], [16], [18]. The system incorporated adversarial training and leveraged model diversity within the ensemble to mitigate these threats. Experimental results demonstrated meaningful improvements in robustness, reducing the success rate of both single-step and iterative adversarial attacks. This reinforces the importance of integrating security-focused strategies when developing AI-powered diagnostic tools for clinical environments.

Although the proposed framework shows promising performance, several opportunities exist for extending this research further. Our experience building machine learning systems for real-world prediction tasks, such as student performance modeling [26], further highlights the importance of robustness and dataset diversity—principles that equally apply to medical imaging applications. Future efforts may explore larger and more diverse MRI datasets collected from multiple clinical centers to improve generalizability across different imaging conditions. Advanced architectures such as Vision Transformers, Swin Transformers, or 3D CNNs may capture richer spatial features and enhance tumor characterization. Additionally, employing stronger or certified adversarial defenses, incorporating explainable AI techniques for clinical transparency, and optimizing the model for deployment on resource-constrained edge devices represent promising directions. Integrating multi-modal patient data—including radiomic features, genomic information, or clinical records—may also lead to more comprehensive diagnostic systems.

In summary, this work establishes a strong foundation for accurate, secure, and robust brain tumor detection using deep learning. By addressing both classification performance and adversarial vulnerabilities, the proposed system moves closer to practical and trustworthy deployment in real-world medical imaging workflows.

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